

AWS API Gateway – Execution Task

Objective

To create an **HTTP API** using AWS API Gateway and integrate it with AWS Lambda to expose a backend service over HTTP.

What is API Gateway?

AWS API Gateway is a managed service that allows users to create, publish, and manage APIs. It acts as a front door for applications to access backend services such as AWS Lambda using HTTP requests.

Use cases :~ API GATEWAY AND LAMBDA configuration

Steps:

Created HTTP API

- API type: **HTTP**
- API name: `lamda-api`
- IP type: IPv4

The screenshot displays the AWS API Gateway console interface for creating a new HTTP API. The breadcrumb navigation at the top reads: **API Gateway** > **APIs** > **Create API** > **Create HTTP API**. On the left, a vertical sidebar shows the four-step process: **Step 1: Configure API** (active), **Step 2 - optional: Configure routes**, **Step 3 - optional: Define stages**, and **Step 4: Review and create**. The main content area is titled **Configure API** and contains the following sections:

- API details**
 - API name:** A text input field containing `lamda-api`. A note states: "An HTTP API must have a name. The name is a non-unique value you use to identify and organize your APIs. To programmatically refer to this API, use the API ID that API Gateway generates for you."
 - IP address type:** Two radio button options: **IPv4** (selected, with a note "Includes only IPv4 addresses.") and **Dualstack** (with a note "Includes IPv4 and IPv6 addresses.")
- Integrations (1)**
 - A dropdown menu shows **Lambda**. A **Remove** button is to its right.
 - AWS Region:** A dropdown menu showing **eu-north-1**.
 - Lambda function:** A search input field containing `arn:aws:lambda:eu-north-1:207662791773:function:hello-lambda` with a clear (X) button.
 - Version:** A dropdown menu showing **2.0** with a **Learn more** link.
 - An **Add integration** button is at the bottom left of this section.

At the bottom right of the console, there are three buttons: **Cancel**, **Review and create** (highlighted in blue), and **Next** (highlighted in orange).

Integrated with Lambda

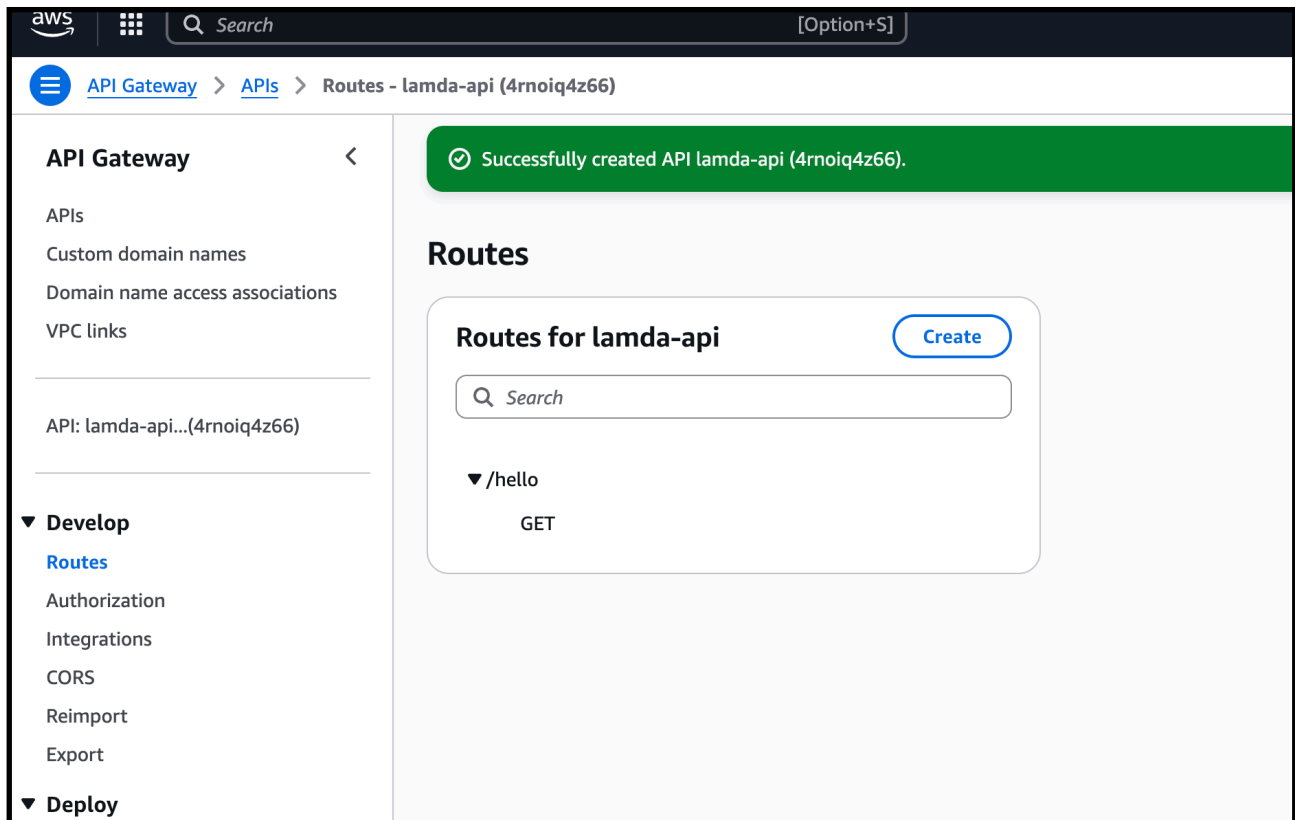
- Lambda function: `hello-lambda`
- Proper permission added automatically
- API Gateway can invoke Lambda

The screenshot shows the 'Configure routes - optional' step in the AWS API Gateway console. On the left, a progress bar indicates the steps: Step 1: Configure API, Step 2: optional: Configure routes (selected), Step 3: optional: Define stages, Step 4: Review and create. The main area is titled 'Configure routes - optional' and contains a 'Configure routes' section with an 'info' icon. Below this, a text box explains that API Gateway uses routes to expose integrations and lists HTTP methods. The configuration fields are: Method (GET), Resource path (/hello), and Integration target (hello-lambda). There is an 'Add route' button and a 'Remove' button. At the bottom right, there are 'Cancel', 'Review and create', 'Previous', and 'Next' buttons.

Created Route

- Route: `GET /hello`
- Method: `GET`
- Resource path working correctly

The screenshot shows the 'Review and create' step in the AWS API Gateway console. The breadcrumb trail at the top is: API Gateway > APIs > Create API > Create HTTP API. The progress bar on the left shows: Step 1: Configure API, Step 2: optional: Configure routes, Step 3: optional: Define stages, Step 4: Review and create (selected). The main area is titled 'Review and create' and contains three sections: 'API name and integrations' (API name: lambda-api, IP address type: IPv4, Integrations: hello-lambda (Lambda)), 'Routes' (Routes: GET /hello -> hello-lambda (Lambda)), and 'Stages' (Stages: \$default (Auto-deploy: enabled)). Each section has an 'Edit' button. At the bottom right, there are 'Cancel', 'Previous', and 'Create' buttons.



Stage Configured

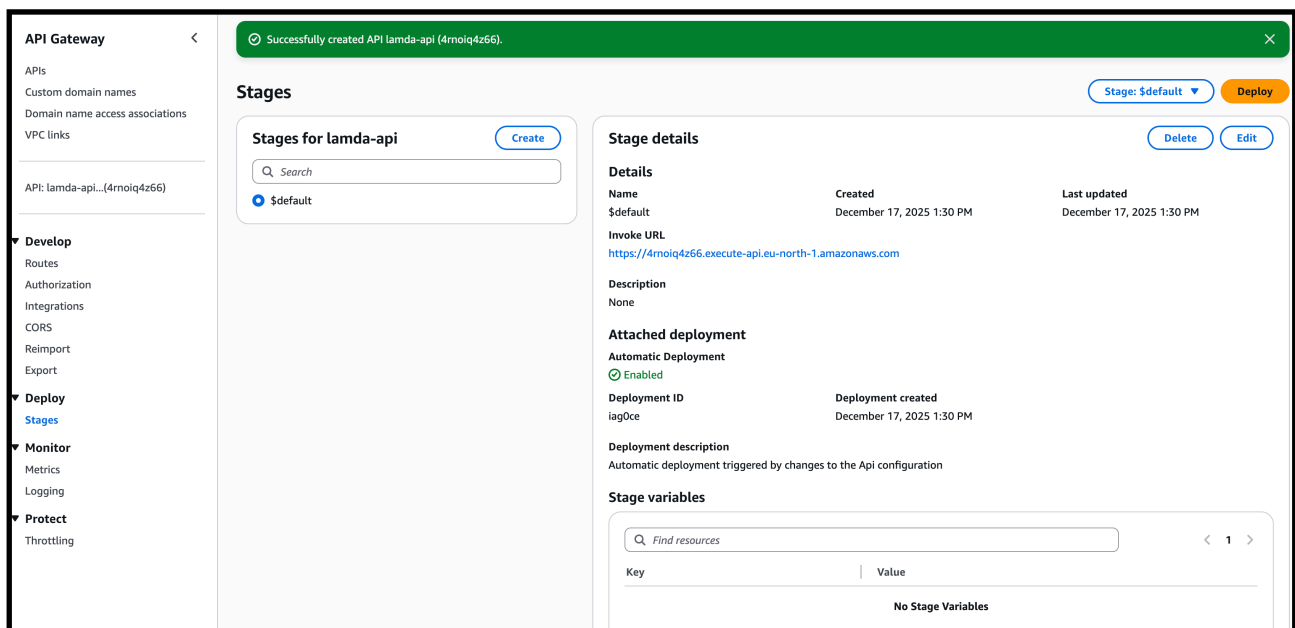
- Stage: \$default
- Auto-deploy enabled

Tested Successfully

- Public endpoint opened in browser
- Output received:

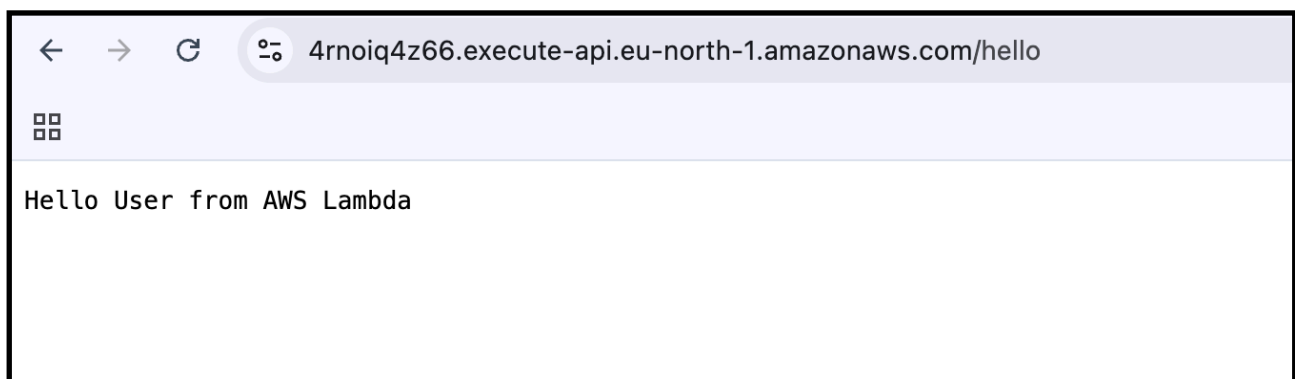
Hello User from AWS Lambda

This confirms end-to-end working.



—> copy the invoke url and paste in browser with /hello

—> [API Endpoint Testing](#)



- This confirms end-to-end working.

Monitoring with CloudWatch

- API Gateway automatically sends execution logs to CloudWatch
- Lambda execution logs are captured separately
- Helps in monitoring request and response flow

Log streams

Tags

Anomaly detection

Metric filters

Subscription filters

Contributor Insights

Data protection

Field

Log streams (14)

By default, we only load the most recent log streams.

Q

Filter log streams or try prefix search

☐ Exact match

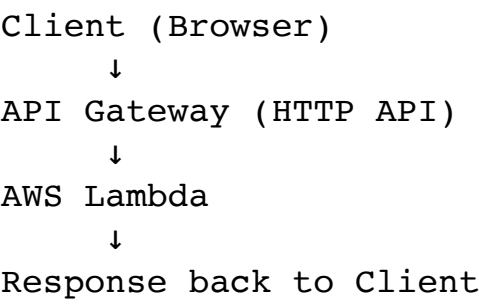
☐ Show expired

[Info](#)

☐	Log stream	Last event time
☐	2025/12/17/[\$LATEST]0566aaf77477420da2f5848006685bed	2025-12-17 11:25:16 (UTC)
☐	2025/12/17/[\$LATEST]c14b82afeb42408392f8857f348cf4b4	2025-12-17 10:45:17 (UTC)
☐	2025/12/17/[\$LATEST]e31a449c7c6348b8bb8b71e66dd8fe4d	2025-12-17 10:44:32 (UTC)
☐	2025/12/17/[\$LATEST]258f2d28aacb47968e171898c9b2debd	2025-12-17 10:26:29 (UTC)
☐	2025/12/17/[\$LATEST]98842be3bf8a4624bd890fa8f40de560	2025-12-17 10:15:30 (UTC)
☐	2025/12/17/[\$LATEST]287dda06163c49a4bfa576b11b15081c	2025-12-17 10:15:25 (UTC)
☐	2025/12/17/[\$LATEST]2c7e79b87f2e413daadd673a57770fc	2025-12-17 10:12:05 (UTC)
☐	2025/12/17/[\$LATEST]70691e406c524097a9d5d7ba3c3e90ec	2025-12-17 10:01:47 (UTC)
☐	2025/12/17/[\$LATEST]059feea7ced24a3e8b9374d46a6f46ff	2025-12-17 09:43:22 (UTC)



Architecture Flow



Key Observations

- API Gateway routes HTTP requests to Lambda
- No server provisioning required
- Supports automatic scaling
- Tight integration with AWS Lambda

